Quantification: 876_z89c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	10
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2006.5	1919.5	977.701	px^2
major_axis_length_px	lacuna shape	80.7401	80.0383	22.7204	px
minor_axis_length_px	lacuna shape	32.0759	34.5853	8.9172	px
aspect_ratio	lacuna shape	2.5567	2.6761	0.4341	unitless
eccentricity	lacuna shape	0.9117	0.9271	0.0379	unitless
solidity	lacuna shape	0.8046	0.8118	0.0648	unitless
roots_count	headline	12	12	5.3748	count
ring_length_r30_px	headline	690.5	716	176.7	px
ring_length_r60_px	reported	1800.5	2064.5	557.017	px
owned_length_px	ownership-dependent	6014.3	6079.75	2900.63	px
edge_count	ownership-dependent	447.4	447.5	217.211	count
mean_edge_length_px	ownership-dependent	13.5133	13.5754	0.4162	px
perimeter_px	normalised	280.598	274.09	95.0149	px
ring_area_r30_px2	normalised	8728.3	8947	2681.26	px^2
ring_area_r60_px2	normalised	21283.9	23194	6687.83	px^2
in_frame_fraction_r30	normalised	0.9799	1	0.0494	unitless
in_frame_fraction_r60	normalised	0.9394	0.9938	0.0929	unitless
ring_density_r30	normalised	0.0810584	0.0797965	0.0118375	px^-1
ring_density_r60	normalised	0.0850804	0.0844376	0.00968498	px^-1
roots_per_100px_perimeter	normalised	4.2267	4.1147	1.0405	per 100 px
ring_attached_length_r30_px	network v2	624.3	692.5	199.591	px
ring_attached_length_r60_px	network v2	1628.3	1816.5	605.801	px
ring_length_w_r30_px	network v2	778.597	815.535	198.343	px
ring_length_w_r60_px	network v2	2039.94	2348.46	634.495	px
sholl_crossings_r10	network v2	15.3	16	4.8546	count
sholl_crossings_r20	network v2	18.5	20	5.9675	count
sholl_crossings_r30	network v2	21.6	23	6.186	count
loop_count	loops, report only	73	73	42.3084	count
loop_edge_share	loops, report only	0.5258	0.5591	0.1497	share
ring_width_mean_r30_px	width	5.9702	5.6859	0.7533	px

Per field
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measure	value	units
analysed_area_px2	1.01891e+06	px^2
total_skeleton_length_px	77906	px
canalicular_length_density_per_px	0.0764598	px^-1
junction_count	2879	count
junction_density_per_px2	0.00282556	px^-2
lacuna_count	13	count

lacunae_per_px2	1.276e-05	px^-2
skeleton_component_count	546	count
mean_component_length_px	142.685	px
median_component_length_px	10	px
field_length_density_w_per_px	0.0861403	px^-1
field_density_without_flagged_per_px	0.0765658	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	90	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0764598	px^-1
field_length_density_with_unattached_per_px	0.0765482	px^-1
loop_count	962	count
loop_edge_share	0.5329	share
edge_count	6110	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2967	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	62934	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2536	138.359	30.0256	4.608	0.9762	0.657	-1.519	12.5193	411.862	True
2	571	54.3172	19.0256	2.855	0.9366	0.633	1.3824	9.958	319.651	True
3	6490	249.288	41.8559	5.9559	0.9858	0.5841	-1.5551	22.8282	886.791	True
4	489	44.533	17.4398	2.5535	0.9201	0.6917	-0.951	40.317	438.83	False
5	3313	115.919	39.5634	2.93	0.94	0.7245	-1.5598	154.062	372.366	False
6	2754	105.554	35.8518	2.9442	0.9406	0.7558	1.5255	165.057	102.925	False
7	3301	102.585	44.3824	2.3114	0.9016	0.8169	-1.4051	307.062	909.902	False
8	2169	78.4025	38.343	2.0448	0.8723	0.8413	1.557	311.008	321.631	False
9	956	61.2609	21.1134	2.9015	0.9387	0.8	1.5273	461.978	61.1475	False
10	2610	87.8807	38.6915	2.2713	0.8979	0.8782	-1.531	535.001	198.764	False
11	1500	59.3512	33.3189	1.7813	0.8276	0.8966	1.5421	545.937	440.713	False
12	1670	81.6741	26.9572	3.0298	0.944	0.8337	-1.4297	692.65	612.224	False
13	1303	70.2403	25.0978	2.7987	0.934	0.8068	-1.3129	914.738	39.4635	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	384	620	625.274	44	14.2108	
2	4	189	501	1028.2	77	13.3532	
3	34	1057	2018	7840.01	569	13.7786	
4	6	292	549	759.701	54	14.0685	
5	23	870	2255	6719.34	491	13.685	5.2883
6	16	862	2155	5440.16	404	13.4658	5.6859

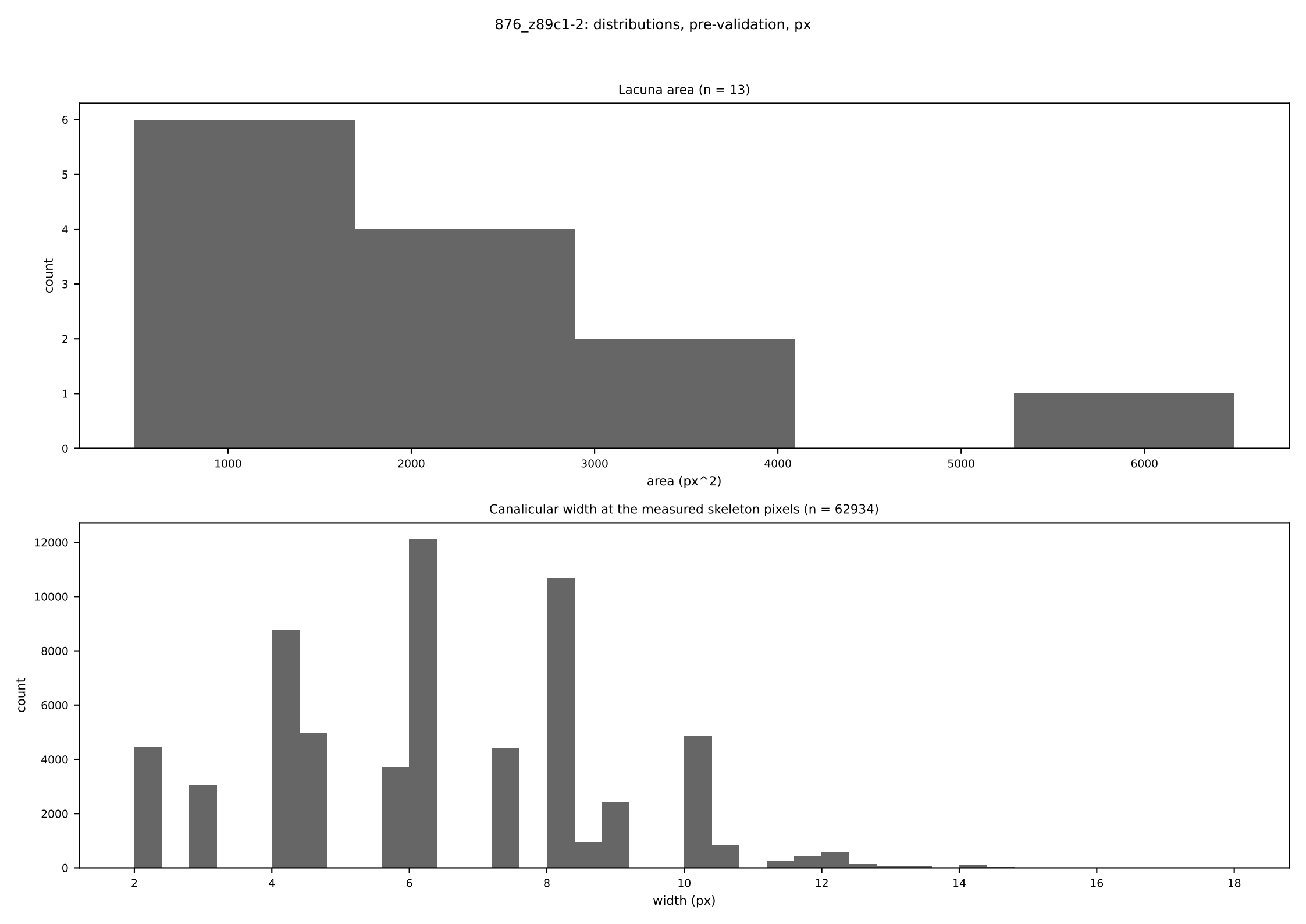
7	16	705	2099	9589.3	699	13.7186	5.4671
8	12	807	2098	5321.55	399	13.3372	6.7128
9	12	724	1641	3740.62	290	12.8987	6.9903
10	13	753	2188	7507.19	575	13.056	6.7179
11	6	708	2031	9832.18	753	13.0573	6.3834
12	9	697	1882	8038.6	581	13.8358	5.6684
13	7	487	1107	3194.32	228	14.0102	4.818

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	583.204	4590	7805	0.6082	0.5704	0.0836601	0.0794363	1.5432
2	192.137	2809	6528	0.6153	0.5722	0.0672837	0.0767463	2.0818
3	1053.68	14114	25792	0.6408	0.5406	0.0748902	0.0782413	3.2268
4	151.995	3280	6332	0.9537	0.8085	0.0890244	0.0867025	3.9475
5	423.014	12381	28256	1	1	0.070269	0.0798061	5.4372
6	413.328	11320	26755	1	0.9628	0.0761484	0.0805457	3.871
7	373.665	11365	27711	1	0.9875	0.0620326	0.0757461	4.2819
8	273.505	9180	23731	1	1	0.0879085	0.0884076	4.3875
9	191.315	7221	16849	1	0.887	0.100263	0.0973945	6.2724
10	277.848	9459	23485	1	1	0.0796067	0.0931658	4.6788
11	190.593	7507	20585	1	1	0.094312	0.0986641	3.1481
12	274.676	8714	22903	1	1	0.0799862	0.0821726	3.2766
13	236.043	6856	16232	0.8455	0.748	0.0710327	0.0681986	2.9656

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	383	521	439.049	703.529	11	10	6
2	183	491	220.995	563.576	5	4	8
3	1037	1910	1178.44	2232.76	29	19	22
4	270	513	334.522	621.375	6	6	10
5	797	2184	966.482	2526.67	19	22	28
6	804	1940	967.767	2438.24	23	23	25
7	532	1693	785.37	2373	14	22	26
8	781	1922	915.683	2378.7	18	20	22
9	685	1550	824.671	1845	18	18	22
10	731	2182	861.754	2505.51	15	20	26
11	700	2008	806.399	2323.93	17	17	22
12	659	1711	783.671	2148.56	13	26	24
13	284	580	539.647	1238.39	10	11	11


876_z89c1-2: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
0
1
2
3
4
5
6
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 62934)
count
width (px)
0
2000
4000
6000
8000
10000
12000
2
4
6
8
10
12
14
16
18